# Lunbing (Caesar) Chen

Bio-inspired Flight · Dynamic Soaring · Aerodynamics · Autonomous Flight

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## Research Profile

I am a Ph.D. candidate in J. C. Wu Center for Aerodynamics at Shanghai Jiao Tong University. My research spans **bio-inspired flight, dynamic soaring, aerodynamics, and autonomous flight**, with a focus on low-energy, long-endurance UAV flight in complex environmental flows. My doctoral research connects two complementary pathways: reducing energy expenditure through passive aerodynamic disturbance mitigation and harvesting atmospheric energy through closed-loop dynamic soaring. My first-author work has been published in the *Journal of Fluid Mechanics* [1], *Cell Reports Physical Science* [2], and *iScience* [3], with a manuscript under external review at *Nature Communications* [4].

## Research Interests

- **Dynamic soaring and autonomous environmental-energy flight**: energy–direction trade-offs, local sensing, state-feedback policies, and closed-loop UAV implementation [3, 4, 8].
- **Bio-inspired aerodynamics and disturbance-resilient flight**: passive disturbance mitigation, avian flight mechanisms, and environmental energy use [2, 9].
- **Unsteady aerodynamics and flow–structure interaction**: aerodynamic sensitivity, vortex dynamics, transient force generation, and reconfigurable structures [1, 5, 6, 10, 11].

## Education

**Shanghai Jiao Tong University** Sep. 2022–Jun. 2027 (expected)
*Ph.D. Candidate, J. C. Wu Center for Aerodynamics* Shanghai, China

- Dissertation topic: *Passive Disturbance Mitigation and Closed-loop Dynamic Soaring for Bio-inspired Long-endurance Flight in Environmental Flows*
- Advisors: Prof. Yang Xiang and Prof. Hong Liu

**Shanghai Jiao Tong University** Sep. 2018–Jun. 2022
*B.Eng., School of Aeronautics and Astronautics* Shanghai, China

## Research Experience

**Mechanics and Feedback Control of Dynamic Soaring** 2022–Present
*Flight mechanics, optimal control, deep reinforcement learning*

- Formulated 3-DOF glider dynamics and a direct-collocation optimal-control framework solved with IPOPT to map flight envelopes and quantify the allocation of harvested energy between sustained soaring and directional progress [3, 8].
- Built a continuous-control reinforcement-learning environment and trained maximum-entropy Soft Actor–Critic policies with curriculum learning and systematic observation/reward ablations to identify the local-sensing and state-feedback requirements of step-level dynamic soaring [4].

**Closed-loop Dynamic Soaring UAV: Integration and Field Testing** 2024–Present
*UAV platform, state estimation, flight control, system integration*

- Designed and integrated a modular sensing–decision–control architecture, combining onboard state and wind sensing, feedback control, safety supervision, and real-time execution on a fixed-wing UAV.

- Built a Gazebo/PX4 simulation and embedded-integration workflow around a Raspberry Pi 5, and conducted field flight tests and manual fixed-wing piloting for platform validation and controller development.

**Bio-inspired Aerodynamics and Passive Flight Robustness** 2022–2026
*Flow–structure interaction, passive reconfiguration, turbulence, aerodynamic sensitivity*

- Designed towing-tank experiments on accelerating reconfigurable plates, combining programmable motion, precision force measurements, high-speed imaging, and time-resolved multi-plane PIV to resolve how fluid–structure interaction redistributes transient loads [1].
- Conducted grid-generated-turbulence wind-tunnel experiments on avian and planform-matched engineered wings using force measurements, hot-wire anemometry, multi-section SPIV, and flight-dynamic analysis to quantify aerodynamic sensitivity, passive stability, and control demand [2].
- Performed laser-sheet smoke visualization and high-speed imaging of live geese during stationary takeoff to connect wing kinematics with the evolving aerodynamic flow structures [9].

**Unsteady Vortex Dynamics and Force Generation** 2021–2025
*Vortex rings, impulse theory, accelerating bodies, PIV, low-order force modeling*

- Combined time-resolved PIV and synchronized force measurements to characterize vortex-ring formation, growth, and wake evolution in accelerating-body flows [1, 10].
- Performed three-dimensional transient simulations in ANSYS Fluent to analyze vortex-ring evolution in confined and asymmetric domains [11].
- Applied the vorticity-moment theorem and impulse-based force decomposition to connect measured vorticity evolution with transient force generation [1].

**Earlier Aerospace Engineering Projects** 2019–2022
*Uncertainty quantification, numerical simulation, engineering software*

- Developed a numerical framework for high-dimensional uncertainty quantification of thermochemical ablation in C/C nozzle throat inserts [7].
- Developed a mobile-based spacecraft management application for status monitoring and task publishing.

## Publications

### First-author / Corresponding-author Publications

*Corresponding author


[1] **Lunbing Chen**, Zhuoqi Li, Qilin Wu, Yufei Yin, Yang Xiang, and Hong Liu. "On the load-shifting phenomenon of accelerating reconfigurable circular plates". In: *Journal of Fluid Mechanics* 1029 (2026), A8. DOI: 10.1017/jfm.2026.11165.

[2] **Lunbing Chen**, Suyang Qin, Qilin Wu, Jinpeng Huang, Yufei Yin, Yong Chen, Yang Xiang, and Hong Liu. "Passive aerodynamic robustness reduces disturbance amplification in flight". In: *Cell Reports Physical Science* 7.10 (2026), p. 103581. DOI: 10.1016/j.xcrp.2026.103581.

[3] **Lunbing Chen**, Yufei Yin, Yang Xiang, Suyang Qin, and Hong Liu. "Optimal dynamic soaring trades off energy harvest and directional flight". In: *iScience* 28.6 (2025). DOI: 10.1016/j.isci.2025.112540.

[4] **Lunbing Chen**, Jixin Lu, Yufei Yin, Jinpeng Huang, Yong Chen, Yang Xiang, and Hong Liu. *Learning step-level dynamic soaring in shear flow*. Under external review at *Nature Communications*. 2026. DOI: 10.48550/arXiv.2604.12413. arXiv: 2604.12413.

[5] Jinpeng Huang, Qilin Wu, **Lunbing Chen***, Yang Xiang, and Hong Liu. "Static Geometric Configuration and Vortex Ring Reshaping Dominate the Load-shifting of Accelerating Plates". Under review at *Physics of Fluids*. 2026.

[6] Qilin Wu**, Lunbing Chen***, Yang Xiang, and Hong Liu. "Shape Independence and Aerodynamic Truncation Effect of Low-Aspect-Ratio Accelerating Plates". In preparation for *Physical Review Fluids*. 2026.

[7] **Lunbing Chen**, Na Sun, Linying Li, and Bin Zhang. "High-Dimensional Uncertainty Quantification of Thermochemical Ablation for C/C Throat Insert". In: *Journal of Propulsion Technology* 43.3 (2022). (in Chinese), pp. 266–275. DOI: 10.13675/j.cnki.tjjs.200583.

### Selected Co-authored Publications

[8] Yufei Yin**, Lunbing Chen**, Yang Xiang, Suyang Qin, and Hong Liu. "Flight Envelope and Energy Equilibrium Strategy of Sailplanes to Perform Dynamic Soaring". In: *International Conference on Aerospace System Science and Engineering*. Springer, 2024, pp. 17–27. DOI: 10.1007/978-981-96-2440-9_3.

[9] Jinpeng Huang, Yang Xiang**, Lunbing Chen**, Suyang Qin, Jixin Lu, Sen Ye, Xiaobin Huang, Yong Chen, and Hong Liu. "Geese achieve stationary takeoff via synergistic wing kinematics and enhanced aerodynamics". Under review in *Science Advances*. 2025. DOI: 10.48550/arXiv.2512.20894.

[10] Zhuoqi Li**, Lunbing Chen**, Yang Xiang, Hong Liu, and Fuxin Wang. "Two independent mechanisms with distinct laws for the generation of drag force on accelerating plates". In: *Physics of Fluids* 36.10 (2024). DOI: 10.1063/5.0227258.

[11] Yufei Yin**, Lunbing Chen**, Liwei Hu, Yang Xiang, Yumin Zhong, and Hong Liu. "Evolution and dynamics of vortex rings in a confined and asymmetric region". In: *Physics of Fluids* 37.12 (2025). **Editor's Pick, *Scilight***. DOI: 10.1063/5.0304637.

[12] Bin Yu**, Lunbing Chen**, Zhen Wang, Xunchen Liu, Bin Zhang, and Hong Liu. "Study on the structural characteristics of strong radial swirl flow based on Lagrangian and Eulerian vortex identification methods". In: *CCTAM 2021*. (in Chinese). 2021.

## Technical Skills

**UAV Systems and Flight Testing**: PX4, Gazebo, Raspberry Pi 5, fixed-wing UAV integration, onboard sensing, manual piloting, field flight testing

**Learning and Optimal Control**: Soft Actor–Critic, curriculum learning, continuous-control simulation environments, direct collocation, IPOPT, trajectory optimization

**Experimental Aerodynamics**: Towing-tank and wind-tunnel experiments, grid-generated turbulence, precision force measurement, PIV/SPIV, hot-wire anemometry, laser-sheet smoke visualization, high-speed imaging

**CFD and Flight Dynamics**: ANSYS Fluent, OpenFOAM, 3/6-DOF flight dynamics, ODE integration, stability analysis, vorticity-moment and impulse methods, low-order modeling

**Scientific Programming**: Python, MATLAB, Fortran, Java, LaTeX, Git